Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Avedisian, C. Thomas Ph.D.		
Project Title:	Influence of Pressure and Composition on Combustion of Sooting Fuel Droplets in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853-7501	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/01/1998	**End Date:**	11/11/2003
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	The broad-based objective of this project is to improve the performance of liquid fuel combustion. The perspective of spherical droplet flames promoted in a microgravity environment is used as the benchmark configuration from which to gain insights on mechanisms and processes for generalization. The project has focussed on three key parameters: ambient pressure, fuel composition and initial droplet diameter on burning. The results have shown the importance of new influences of soot formation that were previously unknown. And results have highlighted the importance of considering detailed flame chemistry and radiation in modeling the measurements. In the current grant year, we performed a number of systematic experiments to examine burning of nonane droplets at high pressure, determined a new scaling that consolidates the burning rates at different initial droplet diameters of nonane, shown the influence of thermal stability additives on combustion of JP8 fuel droplets, and documented the development of soot dynamics at high pressures, up to 15atm, for nonane. The ambient gas is room temperature for all of the experiments. Experiments were performed in both air and in an oxydizing atmosphere consisting of oxygen and helium to compare with a condition where soot formation is minimal. Measurements were made of the droplet, outer luminous zone, and the soot shell diameters using photographic documentation based on color, and black and white, 16mm high speed movie images. The images were digitized and analyzed using commercial software. The broad objective is to develop a detailed understanding of liquid fuel combustion from the perspective of processes at the droplet level with application to environmental air quality and combustion efficiency. Over the past year we carried out several tasks designed to address certain specific questions and processes. These included 1) studying the flame structure and soot dynamics of nonane droplets at elevated pressure; 2) determination of the mechanism responsible for an initial droplet diameter influence on droplet burning, development of a universal correlation for nonane for the droplet diameter range examined, and identification of the possible role of soot formation; and 3) showing how a thermal stability additive blended with JP8 in trace quantities influences the droplet burning dynamics and soot formation. Summaries of these tasks are given in subsequent sections. Many industrial combustion processes are carried out at high pressures, over a wide range of initial droplet diameters representative of spray flames, and for fuels that can be multicomponent, yet little is known about the spherical droplet burning behavior at high pressure and the influence of initial droplet diameter. The challenges are many: creation of spherical droplet flames and soot cloud structures at elevated pressures (greater than 5 atm, for example) is experimentally difficult; development of correlations for droplet burning which can be applied across a range of initial droplet diameters would have great practical value for spray codes that rely on droplet sub-models for the droplet burning rate as a correction to the spherically symmetric value; and determination of the role of thermal stability additives that were developed for processes unrelated to combustion yet which are routinely blended with JP8 and burned as a blend could significantly enhance the development and utilization of such blends if they can be shown to mitigate soot formation. These questions are also very relevant to a terrestrial environment yet we study them from the perspective of spherical droplet flames in microgravity because of the basic and simplified combustion dynamic it promotes, with the droplet and flame being spherical and concentric. This situation corresponds to a one-dimensional transport situation which makes droplet burning modellable and hence valuable for developing the needed understanding for generalization. Spherical droplet flames promoted by microgravity are ideal for examining droplet burning and developing correlations because of the one-dimensional gas phase transport process it promotes which simplifies mechanisms and facilitates understanding. At the same time, results from the spherically symmetric configuration are the limit to which the influences of convection revert as convection is reduced. The dependence of droplet burning rate with droplet diameter and time are among the unique contributions that experimentation in microgravity have revealed. With the current generation of hardware for microgravity experimentation, idealized (surrogate) fuels which are often used as models for multicomponent blends (e.g., the nonane/hexanol mixtures of this study) can give way to practical fuel systems. Doing so further bridges the divide between microgravity and terrestrial applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress provided this period; publication found and added to Bibliography in 2018. See Task Book Archives for previous reporting. https://
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Bae JH, Avedisian CT. "Experimental study of the combustion dynamics of jet fuel droplets with additives in the absence of convection." Combustion and Flame. 2004 Apr;137(1-2):148-62. https://doi.org/10.1016/j.combustflame.2004.02.003 , Apr-2004